

CHAPTER

9

HYPOTHESIS TESTING

Learning Outcomes

On completion of this chapter, the users will be able to

- *Be familiar with the commonly used tests of hypotheses;*
 - *Apply some frequently used hypothesis tests, such as normal test, t-test, chi-squared test and the like;*
 - *Learn about measures of association;*
 - *Gain knowledge on the relative risk and odds ratio.*
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9.1 Preliminaries

Recall that the objective of statistics often is to make inferences about unknown population parameters based on information contained in sample data. These inferences are phrased in one of the two ways: as estimates of the respective parameters or as tests of hypotheses about their values. In many ways the formal procedure for hypothesis testing is similar to the scientific method. The scientist observes nature, formulates a theory, and then tests this theory against observations. In our hypothesis-testing context, the researcher sets up a hypothesis concerning one or more population parameters—that they are equal to some specified values. He then samples the population and compares his observations with the hypothesis. If the observations disagree with the hypothesis, the researcher

rejects it. If not, the researcher concludes either that the hypothesis is true or that the sample failed to detect the differences between the true value and hypothesized value of the population parameters. Examine the following cases:

- A biochemist may wish to determine the sensitivity of a new test for the diagnosis of cancer;
- A production manager asserts that the average number of defective assemblies (not meeting quality standards) produced each Day is 25;
- An Internet server may need to verify if computer users in the country spend on average more than 20 hours on browsing;
- A medical researcher may hypothesize that a new drug is more effective than another in combating a disease;
- An electric engineer may suspect that electricity failure in urban areas is more frequent in rural areas than in urban areas

Statistical hypothesis testing is a procedure is designed to address the above issues with the obtained data. We may now put forward the following definition of a statistical hypothesis.

Definition 9.1: A statistical hypothesis is a statement or assumption regarding one or more population parameters. Our aim in hypothesis testing is to verify whether the hypothesis is true or not on the basis of sample data.

The conventional approach to hypothesis testing is not to construct a single hypothesis, but rather to formulate two different and opposite hypotheses. These hypotheses must be constructed so that if one hypothesis is rejected, the other is accepted and vice versa. These two hypotheses in a statistical test are normally referred to as **null hypothesis** and **alternative hypothesis**.

Definition 9.2: The null hypothesis, denoted by H_0 , is the hypothesis that is to be tested. The alternative hypothesis, denoted by H_1 , is the hypothesis that, in some sense, contradicts the null hypothesis.

Example 9.1: A current area of research interest is the familial aggregation of cardiovascular risk factors in general and lipid levels in particular. Suppose it is known that the average cholesterol level in children is 175 mg/dl. A group of men who have died from heart disease within the past year are identified, and the cholesterol levels of their offspring are measured.

We want to verify if

- The average cholesterol level of these children is 175 mg/dl.
- The average cholesterol level of these children is greater than 175 mg/dl.

This type of question is formulated in a hypothesis-testing framework by specifying the null hypothesis and the alternative hypothesis. In the example above, the null hypothesis is that the average cholesterol level of these children is 175 mg/dl. This is the hypothesis we want to test. The alternative hypothesis is that the average cholesterol level of these children is greater than 175 mg/dl. The underlying hypotheses can be formulated as follows:

Null hypothesis $H_0: \mu = 175$

Alternative hypothesis $H_1: \mu > 175$

We also assume that the underlying distribution is normal under either hypothesis. These hypotheses can be written in more general terms as follows:

Null hypothesis $H_0: \mu = \mu_0$

Alternative hypothesis $H_1: \mu > \mu_0$

In the process of accepting or rejecting a null hypothesis, we may encounter two types of error. We may wrongly reject a true null hypothesis. This leads to an error, which we call **type I error**. The second kind of error, called **type II error**, occurs when we accept a null hypothesis, when it is false, that is, when an alternative is true.

When no error is committed, we arrive at a correct decision. The correct decision may be achieved in two ways: accepting a true null hypothesis or rejecting a false null hypothesis. Four possible outcomes with associated types of error that we commit in our decision are shown in the accompanying table:

Decision	H_0 is true	H_1 is true
Reject H_0	Type I error $P(\text{Type I error}) = \alpha$	Correct decision $P(\text{Correct decision}) = 1 - \beta$
Accept H_0	Correct decision $P(\text{Correct decision}) = 1 - \alpha$	Type II error $P(\text{Type II error}) = \beta$

The probability of committing type I error is usually denoted by α and is commonly referred to as the **level of significance** of a test:

$$\alpha = P(\text{type I error}) = P(\text{rejecting } H_0 \text{ when } H_0 \text{ is true})$$

The probability of committing a type II error is usually denoted by β :

$$\beta = P(\text{type II error}) = P(\text{accepting } H_0 \text{ when } H_1 \text{ is true}).$$

The complement of β , i.e. $1-\beta$ is commonly known as the **power** of a test.

$$1-\beta = 1-P(\text{type II error}) = P(\text{rejecting } H_0 \text{ when } H_1 \text{ is true})$$

Example 9.2: What are the type I and type II errors for the data in Example 9.1?

Solution: The type I error will be committed if we decide that the offspring of men who have died from heart disease have an average cholesterol greater than 170 mg/dl when in fact their average cholesterol level is 175 mg/dl. The type II error will be committed if we decide that the offspring have normal cholesterol levels when in fact their cholesterol levels are above average.

9.1.1 Significance Level

The significance level is the critical probability in choosing between the null and the alternative hypotheses. It is the probability level that is too low to warrant support the null hypothesis. The significance level is customarily expressed as a percentage such as 5% or 1%. A level of significance of say 5% is the probability of rejecting the null hypothesis if it is true. When the hypothesis in question is accepted at the 5% level, the statistician is running the risk that, in the long run, he will be making the wrong decision about 5% of the time.

9.1.2 Test Statistic

The **test statistic** (like an estimator) is a function of the sample observations upon which the statistical decision will be based. The **rejection region (RR)**, specifies the values of the test statistic for which the null hypothesis is **rejected** in favor of the alternative hypothesis. If for a particular sample, the computed value of the test statistic falls in RR, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1 . If the value of the test statistic does not fall into the rejection (critical) region, we

accept H_0 . The region, other than the rejection region, is the **acceptance region**.

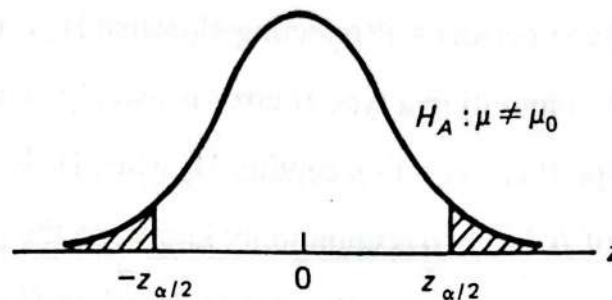


Figure 9.1: Rejection region for two-tailed test for $H_1: \mu \neq \mu_0$

9.1.3 Making Decision

A statistical decision is a decision either to reject or accept the null hypothesis. The decision will depend on whether the computed value of the test statistic falls in the region of rejection or the region of acceptance. If the hypothesis is being tested at 5% level of significance and the observed set of results has probabilities less than 5%, we regard the difference between the sample statistics and the unknown parameter as

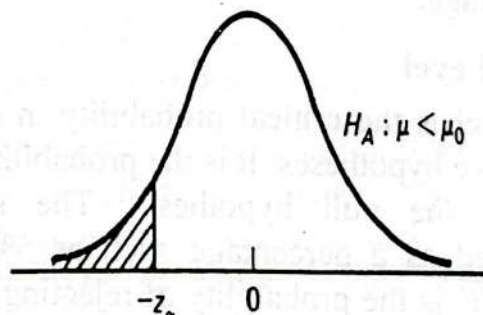


Figure 9.2: Rejection region for one-tailed test for $H_1: \mu < \mu_0$

significant. In other words, we think that sample result is so rare that it cannot be explained by chance variation alone. We then decide to reject the null hypothesis and state that the sample observations are not consistent with the null hypothesis. On the other hand, if at 5% level of

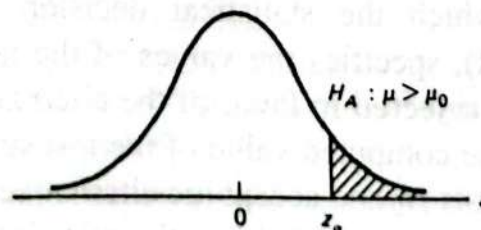


Figure 9.3: Rejection region for one-tailed test for $H_1: \mu > \mu_0$

significance, the observed set of values has probability more than 5%, we give reason that the difference between the sample result and the unknown parameter value can be explained by chance variation and therefore is not statistically significant. Consequently we decide not to reject the null hypothesis and state that the sample observations are not inconsistent with the null hypothesis.

9.1.4 One-tailed and Two-tailed Test

A **one-tailed test** is a test in which the values of the parameter being studied (in our previous example, the mean cholesterol level) under the alternative hypothesis are allowed to be either greater than or less than the values of the parameter under the null hypothesis, but not both. That is we formulate null and alternative hypotheses for a one-tailed test as follows:

$$\begin{array}{ll} \text{Null hypothesis} & H_0: \mu = \mu_0 \\ \text{Alternative hypothesis} & H_1: \mu < \mu_0 \text{ or } \mu > \mu_0 \end{array}$$

A **two-tailed test** is a test in which the values of the parameter being studied under the alternative hypothesis are allowed to be greater than or less than the values of the parameter under the null hypothesis. We formulate the hypotheses under two-tailed test as follows:

$$\begin{array}{ll} \text{Null hypothesis} & H_0: \mu = \mu_0 \\ \text{Alternative hypothesis} & H_1: \mu \neq \mu_0 \end{array}$$

It is very important to realize in a particular application whether we are interested in a one-tailed or two-tailed test.

9.1.5 p -value and its Interpretation

There are two approaches or methods of testing a statistical hypothesis: critical value method and p -value method. The general approach where we compute a test statistic and determine the outcome of a test by comparing the test statistic to a critical value determined by the type I error, is called the **critical-value method** of hypothesis testing. The p -value for any hypothesis test is the alpha (α) level at which we would be indifferent between accepting and rejecting the null hypothesis given the sample data at hand. That is, the p -value is the α level at which the given value of the test statistic (such as t , F , chi-square) would be on the borderline between the acceptance and rejection regions. The p -value can also be thought of as

9.3 The Normal Test

In normal tests, we find a statistic $\hat{\theta}$ whose expected value is specified by the null hypothesis. The standard error of $\hat{\theta}$ is assumed to be known or can be estimated from a large sample. Then we define a variate z as follows:

$$z = \frac{\hat{\theta} - E(\hat{\theta})}{\sigma_{\hat{\theta}}}$$

z is known as a **normal variate** which has mean 0 and variance 1. Any test based on z is commonly known as **normal test**. It is a large sample test and popularly known as z -test. We will illustrate the use of this test by a few examples in the following sections.

9.3.1 Testing the Significance of a Single Mean

Suppose we have a random sample of size n and we wish to test the hypothesis that the population mean μ is equal to some pre-assigned value μ_0 . Two cases will arise here depending on whether the population standard deviation is known or unknown.

Case 1: Population variance known

When population variance σ^2 is known, we can use normal test to compare a sample mean $\bar{x}$ with an assumed population mean μ_0 . Specifically, if $\hat{\theta} = \bar{x}$, the z -test turns out to be a single-mean test under the null hypothesis $H_0: \mu = \mu_0$

$$z = \frac{\bar{x} - \mu_0}{\sigma_{\bar{x}}} = \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma}$$

as our test-statistic.

The critical values of the test-statistic z for two-tailed test when $\alpha = 0.05$ are -1.96 and $+1.96$. The 95% confidence interval for the population mean μ is

$$\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}}$$

Example 9.3: Suppose we assume that cholesterol levels among Bangladeshi women aged 21–40 are approximately normally distributed with mean 190 mg/dl. Blood tests are performed among 200 women of the

same age and the mean level is found to be 181.52 mg/dl. The population standard deviation is known to be 40 mg/dl. What can we conclude with these data? (Use a two-tailed test at 5% level of significance). Compute the 95% confidence interval for the population mean.

Solution: Since the population standard deviation is known, and the sample is also large, we use normal test:

$$z = \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma}$$

The null and the alternative hypotheses are respectively

$$H_0: \mu = 190$$

$$H_1: \mu \neq 190.$$

Hence

$$z = \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} = \frac{\sqrt{200}(181.52 - 190)}{40} = -3.0$$

The critical values are -1.96 and $+1.96$. Because $z = -3.0$, which is less than -1.96 , we can reject the null hypothesis at 5% level of significance.

The 95% confidence interval for the population mean is

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} = 181.52 \pm 1.96 \frac{40}{\sqrt{200}} = (175.98, 187.06)$$

We conclude that in repeated sampling, the true mean will lie in the interval 175.98–187.06 about 95% of the times.

Case 2: Population variance unknown

When the population variance is not known but the sample is large ($n > 30$), we can obtain an estimate of population variance using the sample observations. Consequently, we use $s/\sqrt{n}$ in place of $\sigma/\sqrt{n}$ and the test statistic is written as

$$z = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s}$$

Example 9.4: A random sample of 100 one-quart containers of milk contains a mean of 31.9 ounces with a standard deviation of 0.4 ounces. It is claimed that the containers contain 32 ounces of milk. Test if the

containers are being under-filled at the $\alpha = .01$ level of significance. Compute also the 95% confidence interval.

Solution: The null and the alternative hypotheses are respectively

$$H_0 : \mu = 32 \text{ and } H_1 : \mu < 32$$

since we suspect that the containers actually contain less than the indicated amount of milk. The test is thus one-tailed. The sample values are $n = 100$, $\bar{x} = 31.9$, $s = .4$, so that

$$z = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s} = \frac{\sqrt{100}(31.9 - 32)}{.4} = -2.50$$

From the table in the Appendix, we observe that the rejection region lies to the left of the critical value $z = -2.33$. Hence we reject the null hypothesis and conclude that the containers are apparently being under-filled.

The 99% confidence interval for the mean is

$$\bar{x} \pm 2.33 \frac{s}{\sqrt{n}} = 31.9 \pm 2.33 \left(\frac{.4}{10} \right) = (30.97, 32.83)$$

Example 9.5: A study is conducted to test the claim that the Bangladesh Postal Service delivers letters between Dhaka and other countries of the world within 3 days. A sample of 54 letters is sent and the following delivery times in days were recorded.

1	1	2	2	2	2	2	2	3
3	3	3	3	3	3	3	3	3
3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4
4	4	4	4	4	4	5	5	5
5	5	5	6	6	6	6	8	9

Can we conclude that the postal service achieves its guarantee of a maximum 3-day delivery at the 5% level of significance?

Solution: The null and the alternative hypotheses are

$$H_0 : \mu = 3 \text{ and } H_1 : \mu > 3$$

since the test is one-tailed to the right. Using the sample data

$$\bar{x} = \frac{1+1+2+\dots+6+8+9}{54} = 3.74$$

$$s^2 = \frac{\sum x^2 - n\bar{x}^2}{n-1} = \frac{(1^2+1^2+\dots+8^2+9^2) - 54(3.74^2)}{53} = 2.2801$$

This yields $s=1.51$. Further, since $n=54$ (>30), the distribution of sample mean is approximately normal. Therefore, corresponding to the sample mean $\bar{x} = 3.74$, we find that

$$z = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s} = \frac{\sqrt{54}(3.74 - 3)}{1.51} = 3.60$$

The critical value of z is 1.64 (see table). Consequently, we reject the null hypothesis at the $\alpha = .05$ level of significance. This implies that the mean delivery time for letters between the capital city Dhaka and other parts of the world exceeds 3 days.

The following examples illustrate the use of p -value approach of testing of hypotheses.

Example 9.6: A diary farm study concludes that school aged children consume an average 19.4 ounces of milk per day. In a survey of 140 randomly selected children it is found that they drink an average of 18.5 ounces with a standard deviation of 6.8 ounces. Use p -values to test if the claim is too high at the 1% level of significance.

Solution: The null and the alternative hypotheses are

$$H_0 : \mu = 19.4 \text{ and } H_1 : \mu < 19.4$$

The sample data give

$$\bar{x} = 18.5 \text{ and } s = 6.8 \text{ for } n = 140$$

Since $n > 30$, we employ a normal distribution to find

$$P(\bar{x} \leq 18.5)$$

Corresponding to $\bar{x} = 18.5$, we find that

$$z = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s} = \frac{\sqrt{140}(18.5 - 19.4)}{6.8} = -1.57$$

Referring to a normal integral table in the Appendix

$$P(\bar{x} \leq 18.5) = P(z \leq -1.57) = 1 - P(z < 1.57) = 1 - .9418 = .0582$$

Since this p -value is greater than the level of significance ($\alpha = .01$), we fail to reject the null hypothesis. There is thus insufficient evidence to support the suspicion that the claim is too high.

Example 9.7: The weight of a sample of 40 capsules is to be used to examine whether there is sufficient empirical evidence that the capsules of this type have a mean weight 10 gm. The sample gave a mean weight of 9.77 gm with a standard deviation of .50 gm. Use p -value approach to provide evidence in support of this claim at 5% level of significance.

Solution: The appropriate test here is the normal test with

$$H_0 : \mu = 10 \text{ against } H_1 : \mu \neq 10$$

For $n=40$, the sample values give

$$\bar{x} = 9.77, s = .50$$

With these values

$$z = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s} = \frac{\sqrt{40}(9.77 - 10)}{.5} = -2.91$$

Therefore,

$$P(\bar{x} \leq 9.77) = P(z \leq -2.91) = 1 - P(z \leq 2.91) = 1 - .9982 = .0018$$

Since this is a two-tailed test, we compute p -value by doubling this probability to account for the two tails:

$$p = 2(.0018) = .0036$$

When we compare this p -value to the level of significance desired, we observe that

$$p = .0036 < .05$$

We thus reject the null hypothesis. The mean weight of the capsules is not 10 mg as claimed.

9.3.2 Testing the Equality of Two Population Means

Suppose that the problem is to compare two different groups to see if they represent different populations or if they are essentially equivalent with respect to the mean. To do this, we consider two different populations with

potentially two different means μ_1 and μ_2 . These two populations have standard deviations σ_1 and σ_2 . The null hypothesis in this situation is

$$H_0 : \mu_1 - \mu_2 = 0$$

which asserts that there is no difference or change between the two groups with respect to their means. The alternative hypothesis is often

$$H_1 : \mu_1 - \mu_2 \neq 0$$

so that we have a two-tailed test. Situations sometimes arise involving a one-tailed test where the alternative hypothesis is either

$$\mu_1 - \mu_2 > 0 \text{ or } \mu_1 - \mu_2 < 0$$

To perform such a test, we need two separate independent samples each greater than 30. The test-statistic is

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{x}_1 - \bar{x}_2}}$$

where

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Under the null hypothesis, z follows a normal distribution. Since the null hypothesis asserts that $\mu_1 - \mu_2 = 0$, the formula reduces to

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

It is usually unrealistic to expect that we know the population variances when we do not know the population means. Therefore, if both n_1 and n_2 are larger than 30, we can replace the population standard deviations σ_1 and σ_2 by s_1 and s_2 . This means that for large samples

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \approx \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

The test statistic is thus

$$z \approx \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Example 9.8: A random sample of 71 urban women gave a mean BMI of 22.9 with a standard deviation of 4.2. Another random sample of 83 rural women gave a mean BMI of 18.4 with a standard deviation of 2.2. Test whether there is a difference in the average BMI between the urban and rural women at the 5% level of significance. Compute also the p -value of the test.

Solution: The null hypothesis contends that there is no difference between the average BMI of the urban and the rural women. That is

$$H_0 : \mu_1 - \mu_2 = 0$$

while the alternative hypothesis is

$$H_1 : \mu_1 - \mu_2 \neq 0$$

The sample data provide the following information:

$$n_1 = 71, n_2 = 83, \bar{x}_1 = 22.9, \bar{x}_2 = 18.4, s_1 = 4.2, s_2 = 2.2$$

Since both samples are large ($n_1 > 30, n_2 > 30$), we have

$$\sigma_{\bar{x}_1 - \bar{x}_2} \cong \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{4.2^2}{71} + \frac{2.2^2}{83}} = .55$$

Since this is a two-tailed test at $\alpha = .05$ significance level, the critical values are $z = -1.96$ and $z = 1.96$. Our computed z is

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma_{\bar{x}_1 - \bar{x}_2}} = \frac{22.9 - 18.4}{.55} = 8.18$$

The computed value of z falls in the critical region. We thus reject the null hypothesis and conclude that difference between the BMI of the two groups of women is highly significant at 5% level. The p -value is $p = P(z < -8.18) + P(z > 8.18) = .00$

Example 9.9: A study is conducted among garment's workers to assess if women are discriminated from men in paying the wages in a large city. A random sample of 100 women showed that they are paid a mean hourly wage of Tk. 8.25 with a standard deviation of Tk. 1.25, while another

random sample of 80 men showed that they are paid a mean hourly wage of Tk. 9.50 with a standard deviation of Tk. 2. Do the data provide sufficient evidence that women are paid less than men at the 5% level of significance?

Solution: The null hypothesis is

$$H_0 : \mu_1 - \mu_2 = 0$$

Since our suspicion is that women are paid less, the alternative hypothesis is framed as

$$H_1 : \mu_1 - \mu_2 < 0$$

The sample data give

$$n_1 = 100, n_2 = 80, \bar{x}_1 = 8.25, \bar{x}_2 = 9.50, s_1 = 1.25, s_2 = 2$$

The appropriate test is the normal test with

$$\sigma_{\bar{x}_1 - \bar{x}_2} \approx \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{1.25^2}{100} + \frac{2^2}{80}} = 0.26$$

Since this is a one-tailed test with $\alpha = .05$, the critical value for z is -1.64 .

The computed value of the statistic is

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma_{\bar{x}_1 - \bar{x}_2}} = \frac{8.25 - 9.50}{0.26} = -4.81$$

Clearly, the computed value of z falls in the rejection region. That is women are seemingly paid lower wages, on the average, than men in this sector.

Example 9.10: It has been observed that in Bangladesh, the proportion of women who are delivered through Caesarean sections is very high. A study is therefore conducted to investigate why this is the case. As small height is known to be one of the risk factors related to difficult deliveries, the investigator may want to find out if there is a difference between the mean height of women in Bangladesh who had normal deliveries and of those who had Caesarean sections. The data were as follows:

Type of delivery	Number of women	Mean height in cm	Standard deviation
Normal	60	156	3.1
Caesarean	52	154	2.8

Test the hypothesis that there is no difference between the mean heights of the two groups of women. Use a 5% level of significance.

Solution: Since the sample sizes are large, we employ a normal test and use the sample standard deviations as estimates of population variance. We will test the equality of two population means such that $H_0: \mu_1 = \mu_2$ against the alternative $H_1: \mu_1 \neq \mu_2$ at $\alpha = 0.05$. The critical region is: $z < -1.96$ or $z > 1.96$, where

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

With $\bar{x}_1 = 156$, $\bar{x}_2 = 154$, $n_1 = 60$, $n_2 = 52$, $s_1 = 3.1$, $s_2 = 2.8$, so that

$$z = \frac{156 - 154}{\sqrt{\frac{(3.1)^2}{60} + \frac{(2.8)^2}{52}}} = 3.6$$

The computed value of the statistic z is 3.6, which exceeds the critical value (1.96). Thus the computed p -value is smaller than 0.05., and we reject the null hypothesis and conclude that the observed difference of 2 cm between the mean heights of women with normal deliveries and women with Caesarean sections is a statistically significant difference. Alternately, we can say that the probability that the observed difference of 2 cm between the two groups is due to chance is less than 5 percent.

9.3.3 Testing the Hypothesis for a Specified Value of ρ

Sometimes the correlation between two random variables is expected to be some pre-assigned quantity ρ_0 other than 0 and we wish to test the hypothesis $H_0: \rho = \rho_0$ against the alternative $H_1: \rho \neq \rho_0$. The test is based on Fisher's transformation of r , the sample correlation coefficient. The following example illustrates how this test is performed.

Example 9.11: Suppose the body weights of 100 fathers (x) and first-born sons (y) were measured and the sample correlation coefficient was computed to be 0.38. Verify whether or not this sample correlation is compatible with an underlying correlation of 0.50 that might be expected on genetic ground.

Solution: We want to test the hypothesis $H_0: \rho=0.50$ against the alternative $H_1: \rho \neq 0.50$. The Fisher's transformation of r is given by

$$\lambda = \frac{1}{2} l_n \frac{(1+r)}{(1-r)}$$

The statistic λ is approximately normally distributed under the null hypothesis with mean

$$\xi = \frac{1}{2} l_n \frac{(1+\rho_0)}{(1-\rho_0)}$$

and variance

$$\frac{1}{n-3}$$

The test statistic λ based on the above transformation is a z and is given by

$$z = (\lambda - \xi_0) \sqrt{(n-3)} = \frac{\sqrt{(n-3)}}{2} l_n \left[\frac{(1+r)(1-\rho_0)}{(1-r)(1+\rho_0)} \right]$$

In the present instance, $r=0.38$, $n=100$, $\rho_0=0.50$. Thus

$$\lambda = \frac{\sqrt{(n-3)}}{2} l_n \left[\frac{(1+r)(1-\rho_0)}{(1-r)(1+\rho_0)} \right] = \frac{\sqrt{(100-3)}}{2} l_n \left[\frac{(1+.38)(1-.50)}{(1-.38)(1+.50)} \right] = -1.47$$

The null hypothesis is accepted since our calculated value is greater than the tabulated value (-1.96).

The exact p -value as observed from the normal integral table is

$$p = P(\lambda < -1.47) + P(\lambda > 1.47) = 2[1 - P(z < 1.47)] = 2(.0708) = 0.1416$$

93.4 Testing the Significance of a Single Proportion

The reasoning leading to estimation of one or more means also applies to the problems of estimation of one or more population proportions. We may be interested to infer about the proportion or percentage of smokers in a population, proportion or percentage of children malnourished in population, percentage of people evading income tax and the like. When n elements are randomly sampled from the population, the data will consist of the count x of the number of sampled elements possessing the characteristic. Common sense suggests the sample proportion

$$\hat{p} = \frac{x}{n}$$

as an estimator of the population proportion p .

When the count has the binomial distribution,

$$E(x) = np, \quad V(x) = npq, \text{ where } q = 1 - p$$

Since $\hat{p} = x/n$,

$$E(\hat{p}) = p \text{ and } V(\hat{p}) = pq/n$$

In other words, the sampling distribution of $\hat{p}$ has mean equal to the population proportion. The second result shows that the standard error of the estimator $\hat{p}$ is

$$se(\hat{p}) = \sqrt{\frac{pq}{n}}$$

Consequently, the estimated standard error is

$$se(\hat{p}) = \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

We now consider testing the hypothesis $H_0: p = p_0$ against the alternative $H_1: p \neq p_0$ (also $p > p_0$, or $p < p_0$), and employ the statistic

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0 q_0 / n}}$$

which has standard normal distribution.

Example 9.12: A nutritionist claims that at most 75% of the pre-school children in a certain country have suffered from malnutrition. A sample survey reveals that it is true for 244 children out of a sample of 300 children. Is the nutritionist justified in his claim? Use a significance level of .01.

Solution: The null and the alternative hypothesis are

$$H_0: p = .75 \text{ and } H_1: p > .75$$

The critical value with $\alpha = .01$ is 2.33. Also $\hat{p} = 244/300 = .81$, $p_0 = .75$ Thus

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{.81 - .75}{\sqrt{\frac{.75 \times .25}{300}}} = 2.40$$

Since the computed value of z exceeds 2.33, the null hypothesis is rejected. In other words, we conclude that more than 75% of the children in the country suffer from malnutrition.

Example 9.13: A Consumer Association claims that two-thirds of all children prefer DANO to NIDO. In a sample of 100 children, 55 prefer DANO. Test if the company's claim is overstated at the 5% level of significance.

Solution: The null and the alternative hypotheses are

$$H_0: p = .67 \text{ and } H_1: p < .67$$

which results in a one-tailed test. The sample data give $\hat{p} = .55$. The critical value for z is -1.64 . The test statistic is z defined as

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{.55 - .67}{\sqrt{\frac{.67 \times .33}{100}}} = \frac{-.12}{.047} = -2.55$$

The calculated value of z (-2.55) being smaller than the critical value (-1.64), we reject the null hypothesis and conclude that the Association's claim is overstated at the given level of significance.

Example 9.14: The National Board of Revenue (NBR) asserts that at least 20% taxpayers provide false statement of their actual income while submitting their income tax returns. To check this, the NBR selects 400 returns in their possession and upon investigation, 70 taxpayers were identified to give false statement of their income. What can be said about the NBR's claim? Use 5% level of significance.

Solution: Our hypotheses are

$$H_0: p = 0.20, \quad H_1: p < 0.20$$

Also we have,

$x = 70, n = 400$, so that $\hat{p} = 70/400 = 0.175$ and the critical value is -1.64 .

The computed z -statistic is

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{.175 - .20}{\sqrt{\frac{.2 \times .8}{400}}} = -1.25$$

Because z is greater than the critical value -1.64 , the null hypothesis cannot be rejected at $\alpha = 0.05$. Hence we conclude that the sample data do not provide any evidence that the percentage of taxpayers making false statement in any event is less than 20.

9.3.5 Testing the Equality of Two Population Proportions

We are often interested to compare two populations with regard to the rate of incidence or prevalence of particular characteristic. Let p_1 denote the proportion of members possessing the characteristic in population 1, and p_2 that in population 2. Our goal is to devise a test procedure to test the hypothesis $H_0 : p_1 = p_2$, which says that the two population proportions are equal. The population proportions of successes p_1 and p_2 are estimated by the sample proportions as follows:

$$\hat{p}_1 = \frac{x_1}{n_1}, \text{ and } \hat{p}_2 = \frac{x_2}{n_2},$$

Naturally, $(\hat{p}_1 - \hat{p}_2)$ serves to estimate the difference $(p_1 - p_2)$ and its standard error is given by

$$se(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$$

Under the null hypothesis, we use the following test statistic for large n ,

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where $\hat{p}_1$ and $\hat{p}_2$ are the sample estimates of p_1 and p_2 respectively and $\hat{p}$ is the pooled estimate of p defined as

$$\hat{p} = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2}$$

Example 9.15: A study is undertaken to compare the rates of prevalence of antibody to para-influenza-1 virus among boys and girls in the age group 5 to 9 years. Among 113 boys tested, 34 found to have the antibody.

among 139 girls tested, 54 had the antibody. Do the data provide strong evidence that the rate of prevalence of the antibody is significantly higher in girls than boys? (Use $\alpha = 0.05$). Find also the p -value

Solution: The null hypothesis to be tested is

$$H_0 : p_1 = p_2 \text{ against } H_1 : p_1 < p_2$$

where p_1 is the proportion of boys with antibody and p_2 is the proportion of girls with antibody.

It is easy to find that $\hat{p}_1 = 0.301$, $\hat{p}_2 = 0.388$, $\hat{p} = 0.349$, so that

$$z = \frac{0.301 - 0.388}{\sqrt{0.349 \times 0.651} \sqrt{\frac{1}{113} + \frac{1}{139}}} = -1.44$$

Since the calculated value of z is greater than the critical value (-1.64), we cannot reject the null hypothesis. Consequently, the assertion that the girls have a higher rate of prevalence of antibody than the boys is not substantiated at level $\alpha = 0.05$.

The p -value is

$$p\text{-value} = P(z \leq -1.44) = 0.0749$$

This means that we must allow an α of at least .0749 in order to consider the result significant.

Example 9.16: A random sample of 100 males reveals that 35% of them maintain an account with the national bank. A female sample of the same size revealed that they do so in 40% of the cases. Is this sex difference in maintaining a bank account significant at 5%?

Solution: Here $n_1 = 100$, $n_2 = 100$, $\hat{p}_1 = .35$, $\hat{p}_2 = .40$, so that

$$\hat{p} = \frac{100 \times .35 + 100 \times .40}{100 + 100} = .375$$

If we decide to test tow-tailed hypothesis of no-difference, then

$$H_0 : p_1 = p_2 \text{ and } H_1 : p_1 \neq p_2$$

and the observed z -value is

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{.35 - .40}{\sqrt{.375(1-.375)\left(\frac{1}{100} + \frac{1}{100}\right)}} = -.73$$

The critical value at 5% level is -1.96. Our calculated value lies in the acceptance region, leading us to conclude that there is in fact no sex-difference in maintaining account.

9.3.6 A Quick Guide to Decision Rule for Normal Test

A statistical test may be accomplished at several levels of significance, the most common being 1 percent and 5 percent. We provide a table below to guide you to choose between the null hypothesis and each of the alternative hypotheses at these two levels of significance, viz. $\alpha = .05$ and $\alpha = .01$ in testing the null hypothesis $\theta = \theta_0$.

	Decision rule	
	$\alpha = .05$	$\alpha = .01$
H_1	Reject H_0 if and only if	Reject H_0 if and only if
$\theta > \theta_0$	$z \geq +1.64$	$z \geq +2.33$
$\theta < \theta_0$	$z \leq -1.64$	$z \leq -2.33$
$\theta \neq \theta_0$	$z \geq +1.96$ or $z \leq -1.96$	$z \geq +2.58$ or $z \leq -2.58$

9.4 The t -Test

This test is analogous to normal test except that it is applied when the population variances are unknown and samples are small ($n \leq 30$). We illustrate the use of t -test in the following situations:

- Testing the significance of a single population mean
- Comparing two independent sample means
- Comparing two correlated sample means
- Testing the significance of a zero correlation coefficient

9.4.1 Testing the Significance of a Single Mean

When the sample size is small ($n \leq 30$) and the population variance is unknown, we employ t -test to test the significance of a single population mean. The test statistic is defined as follows:

$$t = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s}$$

where n is the sample size, $\bar{x}$ is the sample mean, μ_0 the specified population mean and s is the sample variance. This is the test statistic that is used to test the null hypothesis $H_0: \mu = \mu_0$ against some suitable alternative. If the null hypothesis is true, the sampling distribution of t is a student t -distribution with $n-1$ degrees of freedom, provided that the underlying population is approximately normal.

Example 9.17: A company finds that a typical office worker spends an average of 13 minutes per hour on non-work related activities. On a trial basis, management sets up new enclosed work stations for a randomly selected group of 12 workers and finds that the average number of minutes lost per hour is 11.3 with a standard deviation of 3.7 minutes. Can we conclude that the new arrangement represents a significant change? Use 5% level of significance.

Solution: The null and the alternative hypotheses are

$$H_0: \mu = 13 \text{ and } H_1: \mu \neq 13$$

We have from the sample data, $n = 12$, $\bar{x} = 11.3$, $s = 3.7$, so that

$$t = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s} = \frac{\sqrt{12}(11.3 - 13)}{3.7} = -1.59$$

This is a two-tailed t -test for 11 df with 2.5% of the population in each tail. As a result, we find that the critical values for t are $t_{0.025} = \pm 2.201$. Comparing the calculated and tabulated t value, we fail to reject the null hypothesis. That is, we cannot conclude that the new work stations produce a significant change in the average lost time per worker.

The 95% confidence interval for the mean is

$$\bar{x} \pm t_{0.025} \left(\frac{s}{\sqrt{n}} \right) = 11.3 \pm 2.201 \left(\frac{3.7}{\sqrt{13}} \right) = (9.04, 13.56)$$

Example 9.18: Suppose the mean cholesterol level of 10 children whose fathers died from heart disease in Example 9.1 is 200 mg/dl and the sample standard deviation is 50 mg/dl. Test the hypothesis that mean cholesterol level is higher in this group than in the general population. Use 5% level of significance.

Solution: It is a one-tailed test and the null and alternative hypotheses are

$$H_0: \mu = 175 \text{ and } H_1: \mu > 175$$

For this test the test-statistic is

$$t = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s}$$

The calculation shows that

$$t = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s} = \frac{\sqrt{10}(200 - 175)}{50} = 1.58$$

The calculated t value then must be compared with the tabulated value at $\alpha = 0.05$ for $10 - 1 = 9$ degrees of freedom. The tabulated value is seen to be 1.833. Our value is less than this. This also implies that the p -value is greater than 0.05. Hence we conclude that our results are not statistically significant and hence the null hypothesis is accepted.

Example 9.19: Test the following hypotheses with the above example:

$$H_0: \mu = 175 \text{ and } H_1: \mu \neq 175$$

Solution: It is a two-tailed test. The tabulated value of t (for $\alpha = 0.05$ and degrees of freedom = 9) is 2.262, which is greater than our computed value of t ($t = 1.58$) as before. Hence we arrive at the same conclusion that our results are not statistically significant and hence the null hypothesis is accepted. Thus the mean cholesterol level of these children is not significantly different from that of an average child.

9.4.2 Testing the Equality of Two Population Means

For small samples, we can base our decision on t-distribution, provided the two populations have approximately normal distribution and equal variances. The null hypothesis is of the form

$$H_0 : \mu_1 = \mu_2$$

The alternative hypotheses may be either of the following form

$$H_1 : \mu_1 > \mu_2 \text{ or } H_1 : \mu_1 < \mu_2$$

Assume that populations from which the two samples of sizes n_1 and n_2 were drawn have means μ_1 and μ_2 with variances σ_1^2 and σ_2^2 respectively. Assume that the underlying variances in the two populations are equal ($\sigma_1^2 = \sigma_2^2 = \sigma^2$) but unknown. The appropriate test-statistic t is defined as

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

which is distributed as t with $n_1 + n_2 - 2$ degrees of freedom.

Since σ is unknown, it must be estimated from the sample as a pooled estimate of σ . This is s , which is to be computed as follows:

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Replacing σ by s

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Example 9.20: Suppose a sample of 8 non-pregnant, pre-menopausal oral contraceptives (OC) users who work on a company are identified on sample basis who have mean systolic blood pressure (SBP) of 132.88 mm Hg and sample standard deviation of 15.34 mm Hg. Another sample of 21 women of similar age who are non-pregnant, pre-menopausal and non-users of OC are similarly identified having mean systolic blood pressure

127.44 mm Hg with sample standard deviation of 18.23 mm Hg. What can be said about the underlying mean blood pressure of these two groups?

Solution: Assume that SBP is normally distributed with mean μ_1 and variance σ_1^2 in the first group and in the second group with mean μ_2 and variance σ_2^2 . Assume that the underlying variances in the two populations are equal ($\sigma_1^2 = \sigma_2^2 = \sigma^2$) but unknown. The appropriate test-statistic t is defined as

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

which is distributed as t with $n_1 + n_2 - 2$ degrees of freedom.

Since σ is unknown, it must be estimated from the sample as a pooled estimate of σ . The pooled variance is estimated first:

$$s^2 = \frac{(8-1)15.34^2 + (21-1)18.23^2}{8+21-2} = 307.18$$

and hence

$$s = \sqrt{307.18} = 17.527$$

Thus under the null hypothesis against the alternative

$$t = \frac{132.88 - 127.44}{17.527 \sqrt{\frac{1}{8} + \frac{1}{21}}} = 0.75$$

For a two-tailed test, the 5% tabulated value of t for $8+21-2=27$ degrees of freedom is 2.052. This value being greater than the calculated value (0.74), it falls in the acceptance region. This leads to the acceptance of our null hypothesis and hence we conclude that the mean blood pressure of the two-groups (OC users and OC non-users) are not significantly different from each other.

9.4.3 Comparing Two Correlated Sample Means: Paired t-test

We have discussed how useful the difference of means test for comparing the means of two different independent groups to determine whether there is a difference between them. However, the test does not apply if the two samples are linked or paired in some fashion, that is, if they are dependent.

Two samples are said to be **paired** or **dependent** when each data point of the first sample is matched and is related to a unique data point of the second sample. Since samples are paired, they must be of equal size. The test statistic is a t and is defined as

$$t = \frac{\sqrt{n}(\bar{d} - \mu_D)}{s_d}$$

which is distributed as student t with $n-1$ degrees of freedom if the differences of the observations in the samples are normally distributed. The null hypothesis is that the mean difference in the population is zero. That is

$$H_0: \mu_D = 0$$

where $\mu_D = \mu_1 - \mu_2 = 0$

The $\bar{d}$ and s are calculated from the sample values as follows

$$\bar{d} = \frac{\sum d_i}{n} \text{ and } s_d = \sqrt{\frac{\sum d_i^2 - n\bar{d}^2}{n-1}}$$

The following examples will illustrate the testing procedure of two dependent samples.

Example 9.21: Suppose systolic blood pressures of 10 women were measured before and after they used OC. The accompanying table shows the level of blood pressure at two points in time.

Women	SBP before using OC (x_{1i})	SBP while using OC (x_{2i})	Difference d_i (Col. 3–Col. 2)
1	115	128	13
2	112	115	3
3	107	106	-1
4	119	128	9
5	115	122	7
6	138	145	7
7	128	132	4
8	105	109	4
9	104	102	-2
10	115	117	2
Total	—	—	46

We are assuming that the underlying mean difference in BP between follow-up and baseline is μ_D . If $\mu_D = 0$, then there is no difference between mean baseline and follow-up BP. If $\mu_D > 0$, then the use of OC pills is associated with an increase in mean BP. If $\mu_D < 0$, then the use of OC pills is associated with a decrease in mean BP. The null hypothesis to be tested is formulated as follows:

$$H_0: \mu_D = 0 \text{ against the alternative } H_1: \mu_D \neq 0$$

The test statistic is t defined under the null hypothesis as

$$t = \frac{\sqrt{n} \bar{d}}{s_d}$$

where $\bar{d}$ = mean of the differences between pairs of observations at two points in time. The test based on t defined above is called **paired t-test**. The t -statistic is distributed as t with $n-1$ degrees of freedom. Using the data in the above table

$$\bar{d} = \frac{13 + 3 + \dots + 2}{10} = 4.6$$

$$\sum d_i^2 = 13^2 + 3^2 + \dots + 2^2 = 398$$

so that

$$s_d = \sqrt{\frac{\sum d_i^2 - n\bar{d}^2}{n-1}} = \sqrt{\frac{398 - 10(4.6^2)}{10-1}} = 4.55$$

Substituting these values in the formula above, we get

$$t = \frac{\sqrt{n} \bar{d}}{s_d} = \frac{\sqrt{10} (4.6)}{4.55} = 3.20$$

The critical value of t at 5% level for 9 degrees of freedom is 2.26. Our computed value of t is greater than this, so we reject the null hypothesis.

Example 9.22: During a nutritional survey, a quality control exercise was carried out to check the agreement between two observers in measuring the children's weight. The results of the survey were as shown in the accompanying table:

Child	Weight (in kg) as measured by		Difference
	Observer I	Observer II	d_i
1	18.6	17.7	0.9
2	17.1	14.5	2.6
3	14.3	12.4	1.9
4	23.2	20.7	2.5
5	18.4	16.8	1.6
6	14.9	14.4	0.5
7	16.6	14.1	2.5
8	14.8	17.1	-2.3
9	21.5	21.2	0.3
10	24.6	21.9	2.7

Can we regard this set of 10 differences as a sample from the population of differences that would have been obtained if the observers had measured the whole population? Use a 5% level of significance to arrive at a decision.

Solution: The null and the alternative hypotheses to be tested are formulated as follows:

$$H_0: \mu_D = 0 \text{ and } H_1: \mu_D \neq 0$$

The test statistic t under the null hypothesis is

$$t = \frac{\sqrt{n} \bar{d}}{s_d}$$

where

$$\bar{d} = \frac{.9 + 2.6 + \dots + .3 + 2.7}{10} = 1.32$$

and

$$s_d = \sqrt{\frac{\sum d_i^2 - n\bar{d}^2}{n-1}} = \sqrt{\frac{39.16 - 10(1.32)^2}{10-1}} = 1.55$$

For $\alpha = .05$ with 9 degrees of freedom the critical values are 2.62 and -2.62 at the right and left tails of the distribution. With the computed values

$$t = \frac{\sqrt{n} \bar{d}}{s_d} = \frac{\sqrt{10} (1.32)}{1.55} = 2.69$$

Since the calculated value of t is greater than the critical values, we reject the null hypothesis. In other words, we can argue that if the observers A and B had measured the weights of all the children in the population, they would have obtained differences in their measurements. The observed differences cannot thus be attributed to the chance factor.

Example 9.23: Suppose that 10 people are randomly selected from a large group of individuals who were given special diets in a weight-loss program. Two sets of measurements were recorded: one before administering the diet and the other one after the diet in a time lag of 6 weeks. The accompanying table displays the data. Determine if the diet results in a significant weight loss at 5% level of significance.

Serial no.	Before (x_i)	After (y_i)	Difference $d_i = (x_i - y_i)$	d_i^2
1	190	185	+5	25
2	202	197	+5	25
3	177	185	-8	64
4	160	152	+8	64
5	225	205	+20	400
6	180	184	-4	16
7	196	185	+11	121
8	208	200	+8	64
9	185	187	-2	4
10	177	170	+7	49
Total	-	-	50	832

Solution: The null hypothesis is

$$H_0: \mu_D = 0$$

Since we want to show that the diet is successful, we want to test whether the mean of all the differences is actually positive to indicate a positive loss. The alternative hypothesis is therefore

$$H_0: \mu_D > 0$$

$$\bar{d} = \frac{5+5+\dots+5-2+7}{10} = 5$$

and

$$s_d = \sqrt{\frac{\sum d_i^2 - n\bar{d}^2}{n-1}} = \sqrt{\frac{832 - 10(25)}{10-1}} = 8.04$$

Since we assumed that the d 's are approximately normally distributed, we can use t -distribution with $n-1$ degrees of freedom. This is a one-tailed test with 5% of the population in the right-hand tail. The critical value of t with 9 degrees of freedom is 1.833. The t -value corresponding to the sample observations is

$$t = \frac{\sqrt{n} \bar{d}}{s_d} = \frac{\sqrt{10}(5)}{8.04} = 1.97$$

which is larger than the critical value (1.833). We reject the null hypothesis and conclude that the diet plan results in weight loss at the 5% level of significance.

9.4.4 Testing the Hypothesis of Zero Correlation

We use the value of the sample correlation coefficient r as an estimate of the population correlation coefficient ρ and perform a statistical test to see if there is any correlation between the variables. In this hypothesis test, the null hypothesis typically asserts that there is no correlation:

$$H_0: \rho = 0$$

while the alternative hypothesis states that

$$H_0: \rho \neq 0$$

so that we have a two-tailed test.

If we suspect that there is only positive correlation or negative correlation, then we could use a one-tailed test and the corresponding alternative hypothesis would be $\rho > 0$, or $\rho < 0$. The test statistic is

$$t = \frac{r\sqrt{(n-2)}}{\sqrt{(1-r^2)}}$$

which is distributed as student's t with $n-2$ degrees of freedom.

Example 9.24: A study is conducted to determine if there is a relationship between the weight of a car and its gas mileage. The following set of

values is obtained and listed in the form (weight in pounds, miles per gallon):

(2800, 19), (2650, 23), (2500, 27), (2450, 25),
(2200, 32), (2300, 26), (2500, 22), (2600, 18)

Determine the coefficient of correlation and test its significance at 5% level of significance.

Solution: To compute the correlation coefficient, we begin with constructing a table as follows:

x	y	xy	x^2	y^2
2800	19	53200	7840000	361
2650	23	60950	7022500	529
2500	27	67500	6250000	729
2450	25	61250	6002500	625
2200	32	70400	4840000	1024
2300	26	59800	5290000	676
2500	22	55000	6250000	484
2600	18	46800	6760000	324

The sums of the column values are

$$\sum x = 20000, \sum y = 192, \sum xy = 474900$$

$$\sum x^2 = 50255000, \sum y^2 = 4752$$

What we need first is to compute the correlation coefficient (r):

$$\begin{aligned} r &= \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}} \\ &= \frac{8(474900) - (20000)(192)}{\sqrt{8(50255000) - 20000^2} \sqrt{8(4752) - 192^2}} \\ &= \frac{-40800}{\sqrt{2040000} \sqrt{1152}} = -.842 \end{aligned}$$

The null and the alternative hypotheses are

$$H_0: \rho = 0 \text{ and } H_1: \rho \neq 0$$

the critical region of t at 5% level of significance for 6 degrees of freedom is $t > 2.447$ or $t < -2.447$. We now compute t -value:

$$t = \frac{-.842\sqrt{(8-2)}}{\sqrt{(1-.709)}} = -3.82$$

The computed value of t being less than the critical value (-2.447), we reject the null hypothesis of zero correlation. In other words, a r value as large as $-.842$ is not likely to occur by chance alone. It is significantly different from zero.

Example 9.25: Suppose blood pressures in 20 randomly selected individuals are measured to determine if there is any correlation between systolic and diastolic blood pressures. Specifically, we wish to test the hypothesis $H_0: \rho=0$ against $H_1: \rho \neq 0$. Suppose that $r=0.58$ for $n=20$. We want to test if this evidence is sufficient to reject the null hypothesis.

Solution: In this instance, the null hypothesis will be rejected if t is sufficiently far from 0. The test statistic is t defined as

$$t = \frac{r\sqrt{(n-2)}}{\sqrt{(1-r^2)}}$$

which, under H_0 , follows t -distribution with $n-2$ degrees of freedom. Substituting the values in the above equation,

$$t = \frac{.58\sqrt{(20-2)}}{\sqrt{(1-.58^2)}} = 3.02$$

For 8 degrees of freedom, with $\alpha=0.05$, the value of t is 2.306. This being smaller than the calculated value, we reject the null hypothesis and conclude that systolic and diastolic blood pressures are related.

9.4.5 Comparing Several Population Means: The F -Test

The statistical method for testing the null hypothesis that the means of several populations are equal is accomplished by what is known as F -test. The F -test is based on the technique of analysis of variance, abbreviated ANOVA. The ANOVA is a very powerful technique of statistical analysis. Although detailed exposition of the technique is beyond the scope of this

text, we shall introduce the readers with the concept of ANOVA for a understanding of the F -test procedure. The term **analysis of variance** describes a technique whereby the total variation embedded in the data being analyzed or divided into meaningful components due to independent causes. In the present context, consideration is given to the **one-way analysis of variance**, in which only one type of treatment or factor is involved in the experiment.

Assume that we have k independent samples from k normal populations with means $\mu_1, \mu_2, \dots, \mu_k$ respectively with common variance σ^2 . The k different populations are often classified according to different treatment or groups. We wish to derive an appropriate method for testing the hypotheses of the form

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k.$$

$$H_1: \text{At least two of the means are not equal}$$

The total sum of squares (SS_T) measuring the total variability in the data can be partitioned and resolved into **between-class sum of squares** (SS_B) and **within-class sum of squares** (SS_W). Expressed symbolically, the sum of squares identity is

$$SS_T = SS_B + SS_W$$

In practice, one usually computes SS_T and SS_B and then using the sum of squares identity, obtain

$$SS_W = SS_T - SS_B$$

The degrees of freedom associated with SS_T is $N-1$ and that with SS_B is $k-1$. Since the total degrees of freedom is $N-1$, the SS_W degrees of freedom is $N-k$. If we divide the SS_B and SS_W by their respective degrees of freedom, we obtain what we call **between-class mean squares** MS_B and **within-class mean squares** MS_W . Symbolically

$$MS_B = \frac{SS_B}{k-1} \text{ and } MS_W = \frac{SS_W}{N-k}$$

Having defined the above quantities, the test statistic to test the null hypothesis is the F -ratio defined as follows:

$$F = \frac{MS_B}{MS_W}$$

which follows F distribution with $\nu_1 = k-1$ and $\nu_2 = N-k$ degrees of freedom. The null hypothesis is to be rejected if $F > F_\alpha$, where F_α lies in the upper tail of the F -distribution with $\nu_1 = k-1$ and $\nu_2 = N-k$ degrees of freedom.

The computations in an analysis of variance problem are usually summarized in tabular form as follows:

Summary table for one-way variance table for computing F

Source of variation	Degrees of freedom	Sum of squares	Mean squares	Computed F
Between classes	$k-1$	SS_B	$MS_b = \frac{SS_B}{k-1}$	$\frac{MS_B}{MS_w}$
Within classes	$N-k$	SS_W	$MS_w = \frac{SS_W}{N-k}$	-
Total	$N-1$	SS_T	-	-

Example 9.26: Suppose we wish to determine whether the starting salaries of secondary school teachers employed in government, non-government and NGO-managed schools of Bangladesh are significantly different. That is we wish to test the null hypothesis that the three population means are the same against the alternative that they are different. For the sake of simplicity, we assume that the three samples are of equal size. The three randomly selected independent samples yield the following results (monthly salaries in hundreds of taka)

	Government	Non-government	NGO-managed
	56	48	55
	60	61	60
	50	48	44
	65	52	46
	64	46	55
Total	$T_1=295$	$T_2=255$	$T_3=260$

Solution: The null hypothesis to be tested is that the average starting salaries of the three categories of teachers are equal against the alternative that at least two categories of teachers have different average salaries.

The various quantities required for computing the F -ratio are as follows:

$$\begin{array}{llll} n_1=5 & n_2=5 & N_3=5 & N=15 \\ T_1=295 & T_2=255 & T_3=260 & G=810 \\ T_1^2=87,025 & T_2^2=65,025 & T_3^2=67,600 & G^2=656,100 \end{array}$$

$$SS_T = 56^2 + 60^2 + \dots + 55^2 - \frac{810^2}{15} = 44408 - 43740 = 668$$

$$SS_B = \frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \frac{T_3^2}{n_3} - \frac{G^2}{N} = \frac{87025}{5} + \frac{65025}{5} + \frac{67600}{5} - \frac{656100}{15} = 190$$

$$SS_W = SS_T - SS_B = 668 - 190 = 478, \quad k = 3$$

The ANOVA table is constructed with the values computed above:

Source of variation	df	Sum of squares	Mean squares	Computed F
Between classes	$\nu_1=2$	190	$\frac{190}{2} = 95$	$\frac{95}{39.8} = 2.4$
Within classes	$\nu_2=12$	478	$\frac{478}{12} = 39.8$	-
Total	14	668	-	-

The critical value of F -ratio that cuts off the upper 5 percent of the distribution with $\nu_1=2$ and $\nu_2=12$ degrees of freedom is 3.88. This is greater than the computed F . Thus at $\alpha=0.05$, we conclude that no significant difference exists in starting salaries of secondary school teachers in the categories of schools.

Example 9.27: In a study on the effect of diet on sugar in the blood, 32 subjects were selected for their uniformity and assigned randomly to four groups, eight cases being assigned to each group. A mishap resulted in the loss of the records for six cases. The following data represent the results of the remaining cases.

Diet			
I	II	III	IV
24	26	30	30
18	21	32	28
25	23	29	27
23	25	25	23

Diet			
I	II	III	IV
22	20	31	31
-	24	33	25
-	20	29	-
-	-	28	-
$T_1=112$	$T_2=159$	$T_3=237$	$T_4=164$

Test at 5% level the hypothesis that the diets produce equal mean effects on sugar in the blood.

Solution: The null hypothesis to be tested is that the four groups produce equal mean effects on sugar in the blood against the alternative that at least two groups produce unequal mean effects on sugar in the blood.

The required quantities for the ANOVA are

$$SS_T = 24^2 + 18^2 + \dots + 25^2 - \frac{672^2}{26} = 409.385$$

$$SS_B = \frac{112^2}{5} + \frac{159^2}{7} + \frac{237^2}{8} + \frac{164^2}{6} - \frac{672^2}{26} = 255.548$$

$$SS_W = SS_T - SS_B = 409.385 - 255.548 = 153.837, \quad k = 4$$

The ANOVA table appears below:

Source of variation	df	Sum of squares	Mean squares	Computed F
Between classes	$\nu_1=3$	255.548	$\frac{255.548}{3} = 85$	$\frac{85.18}{6.99} = 12.18$
Within classes	$\nu_2=22$	153.837	$\frac{153.837}{22} = 6.99$	-
Total	25	409.385	-	-

Since the observed F -ratio (12.18) exceeds the critical F -value (3.05) for $\nu_1=3$ and $\nu_2=22$ degrees of freedom, we reject the null hypothesis that the diets produce equal mean effects on sugar in the blood.

9.5 Chi-Square and Chi-Square Based Tests

The chi-square test is one of the most commonly used tests in statistics. Generally, it is used for testing hypotheses concerning the distribution of a random variable rather than the parameter (μ or σ) of the distribution, for which it is referred to as **nonparametric test**. The hypothesis may be regarding:

- a) the distribution of a discrete variable (e.g. the number of children in a family who have measles has a binomial distribution)
- b) the distribution of a continuous variable (e.g. the heights of pregnant mothers have a normal distribution)
- c) the independence of two or more random variables (e.g. the blood group distribution in a population is independent of sex).

We shall illustrate the use of the chi-square for testing the hypotheses concerning the independence or in other words association between two attributes. By very nature of the data, the chi-square test treated here falls under non parametric test. The formula is as follows:

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad (a)$$

where O denotes the observed cell frequency and E the expected cell frequency in a contingency table. The formula can be further simplified as below:

$$\chi^2 = \sum \frac{O^2}{E} - n \quad (b)$$

where n is the total number of observations (or sample size) in a $r \times c$ contingency table. The value for the chi-square statistic is compared to the associated critical value based on the degrees of freedom determined as

$$df = (r - 1)(c - 1)$$

The expected cell frequencies E_{ij} in a contingency table can be calculated as follows:

$$\text{Expected cell frequency} = \frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$$

Let us illustrate the testing procedure by an example.

Example 9.28: To determine the possible effect of a chemical treatment on the rate of seed germination, 100 chemically treated seeds and 150 untreated seeds are sown. The numbers of seeds that germinate are displayed in the accompanying table. Do the data provide strong evidence that the rate of germination is different for the treated and untreated seeds?

	Germinated	Not germinated	Total
Treated	84	16	100
Untreated	132	18	150
Total	216	34	250

Solution: The expected frequencies E_{11} , E_{12} , E_{21} and E_{22} corresponding to 84, 16, 132, and 18 are respectively

$$E_{11} = \frac{100 \times 216}{250} = 86.4, \quad E_{12} = \frac{100 \times 34}{250} = 13.6$$

$$E_{21} = \frac{150 \times 216}{250} = 129.6, \quad E_{22} = \frac{150 \times 34}{250} = 20.4.$$

Applying (a) formula

$$\chi^2 = \frac{(84 - 86.4)^2}{86.4} + \frac{(16 - 13.6)^2}{13.6} + \frac{(132 - 129.6)^2}{129.6} + \frac{(18 - 20.4)^2}{20.4} = 0.82$$

Formula (b) yields

$$\chi^2 = \frac{84^2}{86.4} + \frac{16^2}{13.6} + \frac{132^2}{129.6} + \frac{18^2}{20.4} - 250 = 0.82, \text{ as before}$$

At 5% level of significance, the tabulated value of the chi-square for 1 degree of freedom is 3.841. This being less than our calculated value, we accept the null hypothesis. In other words, chemical treatment makes no difference in the germination of seeds.

Example 9.29: The blood type distribution of a human population is known to vary with national origin. The data in the accompanying table are adapted from a book entitled *Immunology and Serology* by P.L. Carpenter.

Blood type	US	Chinese	Norwegian	Total
O	450	300	190	940
A	410	250	250	910
B	100	150	40	490
AB	40	350	20	160
Total	1000	1000	500	2500

Test the null hypothesis that the blood type distribution is independent of the national origin.

Solution: To apply chi-square test we first compute the expected frequencies

$$E_{11} = \frac{940 \times 1000}{2500} = 376, E_{12} = \frac{940 \times 1000}{2500} = 376, E_{13} = \frac{940 \times 500}{2500} = 188,$$

$$E_{21} = \frac{910 \times 1000}{2500} = 364, E_{22} = \frac{910 \times 1000}{2500} = 364, E_{23} = \frac{910 \times 500}{2500} = 364,$$

$$E_{31} = \frac{490 \times 1000}{2500} = 196, E_{32} = \frac{490 \times 1000}{2500} = 196, E_{33} = \frac{490 \times 500}{2500} = 98,$$

$$E_{41} = \frac{160 \times 1000}{2500} = 64, E_{42} = \frac{160 \times 1000}{2500} = 64, E_{43} = \frac{160 \times 500}{2500} = 32$$

The chi-square value is thus

$$\chi^2 = \sum \frac{(O - E)^2}{E} = \frac{(450 - 376)^2}{376} + \dots + \frac{(20 - 32)^2}{32} = 333.0$$

In this instance, $r = 4$ and $c = 3$, so that the chi-square has 6 degrees of freedom. The critical value of chi-square at 1% level of significance is 16.81. The observed chi-square value is highly significant. We reject the null hypothesis of no association and conclude that blood type depends on the national origin.

An important special case of the two-way contingency table emerges when both variables are dichotomous. The resulting table is a 2 by 2 table or a four-fold table. The accompanying table is a typical one of this form.

Categories of X	Categories of Y		Row totals
	1	2	
1	n_{11}	n_{12}	m_1
2	n_{21}	n_{22}	m_2
Column totals	n_1	n_2	n

The formula for chi-square for such a table can be written as

$$\chi^2 = \frac{n(n_{11}n_{22} - n_{12}n_{21})^2}{m_1 m_2 n_1 n_2}$$

Example 9.30: Suppose a study is conducted in order to evaluate the effectiveness of a new drug (D_1) compared to an old drug (D_2) for relieving hay fever symptoms. A group of 200 patients receives the new drug and another 200 patients receive the old drug. A total of 170 patients who received the new drug, find the drug effective, while 150 patients who received the old drug, find the same effective. What can be said about the effectiveness of the new drug? The data are displayed in table below.

Responses	Drug type		Row totals
	D_1	D_2	
Effective	170	150	320
Not effective	30	50	80
Column totals	200	200	400

The computed chi-square is

$$\chi^2 = \frac{400(170 \times 50 - 150 \times 30)^2}{320 \times 80 \times 200 \times 200} = 6.25$$

Since 6.25 exceeds the critical chi-square value 3.84 associated with chi-square for 1 degree of freedom at 5% level of significance, we reject the null hypothesis and conclude that two drugs are not equally effective,

Example 9.31: Consider a group of 800 children divided according to sex and congenital abnormality. The data are shown in the accompanying table. Test if sex composition influences the congenital abnormality.

Congenital abnormality	Sex		Row totals
	Boy	Girl	
Abnormal	70	50	120
Normal	330	350	680
Column totals	400	400	800

Solution: The chi square value is

$$\chi^2 = \frac{800(70 \times 350 - 50 \times 330)^2}{120 \times 680 \times 400 \times 400} = 3.92$$

The 5% tabulated value of chi-square for 1 degree of freedom is 3.841. Our calculated value is 3.92, which exceeds the tabulated value at 5% level. We thus reject the null hypothesis of independence and conclude that sex composition has bearing upon the congenital abnormality.

Example 9.32: Suppose that in a study of the factors affecting the utilization of antenatal clinics, a researcher found that 64% of the women who lived within 10 kilometers of the of the clinic came for antenatal care (ANC), compared to only 47% of those who lived more than 10 kilometers away. This suggests that ANC is used more frequently by women who live close to the center. The complete results are shown below:

Distance	Used ANC	Didn't use ANC	Total
Less than 10 km	51 (64%)	29 (36%)	80 (100%)
10 km or more	35 (47%)	40 (53%)	75 (100%)
Total	86	69	155

Test at 5% level of significance whether the difference so observed in utilization of antenatal care between those who live close to the centre and those who live far away from the clinic is statistically significant.

Solution: Here $n=155$, $m_1=80$, $m_2=75$, $n_1=86$, $n_2=69$. Hence

$$\chi^2 = \frac{155(51 \times 40 - 29 \times 35)^2}{(80)(86)(75)(69)} = \frac{162846870}{35604000} = 4.57$$

The critical value of the test statistic at 5% level of significance is 3.841. Our observed value 4.57 exceeds the critical value, which implies that the observed p -value is smaller than .05. The null hypothesis is thus rejected in favor of the alternative. We conclude that the women living within a distance of 10 km from the clinic use antenatal care services significantly more often than the women living more than 10 km do.

The exact p -value of the test statistic seems to lie between $\chi^2_{.05} = 3.841$ and $\chi^2_{.01} = 6.635$. Interpolating linearly between these two, we find that p -value=.035, which is less than .05 (i.e. $p<.05$).

9.5.1 Yate's Continuity Correction in Chi-Square Statistic

The statistic on which we base our decision has a distribution that is only approximated by the chi-square distribution, which is continuous. The computed chi-square values depend on the cell frequencies and consequently are discrete. The continuous chi-square distribution seems to approximate the discrete sampling distribution very well, provided the number of degrees of freedom is greater than 1. In a 2×2 contingency table, where we have only 1 degree of freedom a correction called *Yates' correction for continuity* is thus necessary. This correction is analogous the continuity correction for normal approximation to the binomial. The corrected χ^2 value is computed as follows:

$$\chi_c^2 = \frac{N \left(|ad - bc| - \frac{N}{2} \right)^2}{(a+b)(a+c)(c+d)(b+d)}$$

The correction is used, however, only when the number of degrees of freedom is 1.

When the continuity correction is applied to the data in Example 9.32, the computed χ_c^2 is

$$\chi_c^2 = \frac{155 \left(|2040 - 1015| - \frac{155}{2} \right)^2}{80 \times 86 \times 75 \times 69} = \frac{155 \times 947.5^2}{35604000} = 3.91.$$

which is smaller than the one obtained above before continuity correction but the result still remains significant at 5% level.

9.5.2 Mantel-Haenzel χ^2 Test

Sometimes a variable is found to be associated with the dependent variable (the problem/effect) and with the independent variable (the cause of the problem). Such a variable is called **confounding variable**. This may either strengthen or weaken the apparent relationship between the problem and a possible cause. A relationship may exist, for example, between the level of mother's education and the prevalence of malnutrition in under-5 children. However, a third variable, family income may be related to malnutrition. To give a true picture of cause and effect, the confounding variables must be considered in all statistical applications. We illustrate below by an example how this problem is handled in hypothesis testing when association of attributes is in consideration.

Example 9.33: The accompanying table reports the results of a nutrition survey in two communities among the illiterate and literate mothers.

(a) Prevalence of malnourishment in children of all mothers

	Community A	Community B	Total
Malnourished	80 (32%)	80 (32%)	160 (32%)
Not malnourished	170 (68%)	170 (68%)	340 (68%)
Total	250 (100%)	250 (100%)	500

The results of the investigation reveal that the prevalence of malnutrition is the same in both communities (32%). The researcher, however, suspects that mother's education is a confounding variable which might suppress the true relationship. He therefore splits up the table into two as below that shows the prevalence by level of education also.

(b) Prevalence of malnourishment in children of illiterate mothers

	Community A	Community B	Total
Malnourished	37 (62%)	73 (38%)	110 (44%)
Not malnourished	23 (38%)	117 (62%)	140 (56%)
Total	60 (100%)	190 (100%)	250

$$\chi^2 = 9.08, p < .01$$

(c) : Prevalence of malnourishment in children of literate mothers

	Community A	Community B	Total
Malnourished	43 (23%)	7 (12%)	50 (20%)
Not malnourished	147 (77%)	53 (88%)	200 (80%)
Total	190 (100%)	60 (100%)	250

$$\chi^2 = 2.78, p > .05$$

If we examine tables 2 and 3, we observe that

- Within each level of education, malnutrition is more prevalent in community A than in community B (62% Vs 38% and 23% Vs 12%).
- Malnutrition is more prevalent in children of illiterate mothers than in children of literate mothers (44% vs 20%).

Here level of education is a confounding variable because it is related to the variable of interest (malnutrition) and to the groups being compared. It is therefore important to analyze the data for different education levels separately. The appropriate test statistic is the chi-square. This test shows that the difference is significant for children of illiterate mothers ($p < .01$) but not for those with literate mothers ($p > .05$).

Another useful way is to perform a summary test which pools the evidence from the individual tables, but takes into account the confounding factor (here literacy level). This is the so called Mantel-Haenszel χ^2 test, which we describe below.

For a 2 by 2 contingency table, the test statistic is defined as

$$\chi_{MH}^2 = \frac{(n_{11} - E_{11} - .5)^2}{V(n_{11})}$$

Here n_{11} denotes the observed value, E_{11} , the corresponding expected value, and $V(n_{11})$ is the variance of n_{11} .

For the illiterate group

$$n_{11} = 37, E_{11} = \frac{m_1 \times n_i}{n} = \frac{110 \times 60}{250} = 26.4$$

$$V(n_{11}) = \frac{m_1 \times m_2 \times n_1 \times n_2}{n^2(n-1)} = \frac{110 \times 140 \times 60 \times 190}{250^2(250-1)} = 11.28$$

For the literate group

$$n_{11} = 43, E_{11} = \frac{m_1 \times n_i}{n} = \frac{50 \times 190}{250} = 38.$$

$$V(n_{11}) = \frac{m_1 \times m_2 \times n_1 \times n_2}{n^2(n-1)} = \frac{50 \times 200 \times 190 \times 60}{250^2(250-1)} = 7.32$$

For the two tables together

$$n_{11} = 37 + 43 = 80, E_{11} = 26.4 + 38 = 64.4 \text{ and } V(n_{11}) = 11.28 + 7.32 = 18.6$$

Hence

$$\chi^2_{MH} = \frac{(n_{11} - E_{11} - .5)^2}{V(n_{11})} = \frac{(80 - 64.4 - .5)^2}{18.6} = 12.2$$

For 1 degree of freedom, the 5% value of chi-square is 3.841. Thus our result is highly significant (in fact $p < .001$). It can therefore be concluded that the prevalence of malnutrition is significantly different in A and B. Note that this seemed not to be the case when we examined the data in the first table, in which the data for both literate and illiterate mothers were pooled.

9.5.3 McNemar's χ^2 Test

The McNemar's test may be used with either nominal or ordinal data and is especially useful with before-after measurement of the same subject or when both members of a pair give the same response. The pair of responses may result from comparing answers for the same individual at two times, or from comparison of how an individual answers two different questions. The layout of the table for accomplishing this test with the first and second set of responses is as follows:

	After	
	Favor	Do not favor
Before		
Favor	<i>a</i>	<i>b</i>
Do not favor	<i>c</i>	<i>d</i>

Since $b+c$ represents the total number of cases who changed (a and d are no-change responses), the expectation under a null hypothesis is that $\frac{1}{2}(b+c)$ cases change in one direction and the same proportion in the other direction. The McNemar's test uses the following version of the chi-squared test:

$$\chi^2 = \frac{(|b-c|-1)^2}{b+c} \text{ with d.f.}=1$$

A word of caution is necessary here. This test is valid only when $(b+c)$ is larger than 10.

Example 9.34: The accompanying table shows the results of a case-control study that was conducted to determine causes of a cholera outbreak in Italy. For each cholera case confirmed in the hospital, a subject was sought of the same sex, the same age, and the same neighborhood. Test at 5% level of significance if the sample data provide any evidence to indicate that sea food is a risk factor for the outbreak of cholera.

	Cases		
Controls	Ate seafood	Did not eat seafood	Total
Ate seafood	12	3	15
Did not eat seafood	30	31	61
Total	42	34	76

We interpret the table as follows: In 12 pairs both cases and controls ate seafood and in 31 pairs neither of them ate seafood. These 43 pairs therefore give no information about whether eating seafood is a risk factor for getting cholera. However, in 30 pairs the cases ate seafood, while the control did not, whereas in only 3 pairs the controls ate seafood, while the cases did not. Apparently, it seems that eating seafood was a risk factor for the outbreak of cholera. To be statistically confirmed, we employ McNemar's chi-square test with $b=3$ and $c=30$ under the null hypothesis that proportion of controls eating sea fishes is the same as the proportion of cases eating sea fishes. Hence

$$\chi^2 = \frac{(|3 - 30| - 1)^2}{3 + 30} = 20.5$$

This value is highly significant ($p < .01$). We therefore reject the null hypothesis (H_0 : there is no real association between eating seafood and getting cholera) and conclude that eating seafood is a risk factor for getting cholera.

The method is further illustrated by another example in which each individual answers two different questions

Example 9.35: In a survey general practitioners (GP) were asked about the importance of screening mammography (SM) and clinical examination (CE) of the breasts. Of the 194 who responded to both questions, 184 (95%) agreed that CE was important, while only 146 (75%) agreed with the importance of SM. The full information is provided in the

accompanying table. Test if there is any evidence to support the hypothesis that the clinical examination is important to screening mammography.

	Clinical Examination		Total
	Important	Not important	
Screening mammography			
Important	144	2	146
Not important	40	8	48
Total	184	10	194

Solution: The null hypothesis is that the proportion of GPs who agree that CE is important is the same as the proportion who agree that screening mammography is important.

The 144 GPs who agree that both are important have cancelled out, and the 8 who think neither is important have not featured. The significance of the difference between the two proportions depends on the number of **untied** or **discordant** pairs. Thus we see that for such data McNamer's test is applicable.

$$\chi^2 = \frac{(|2 - 40| - 1)^2}{2 + 40} = 32.6 \text{ with 1 d.f.}$$

When compared with the tabulated value of chi-squared, we find the calculated value to be highly significant. Hence there is a strong evidence that more GPs think that clinical examination is important (95%).

In the above example if we let p_1 to denote the proportion who agree that clinical examination is important, then

$$p_1 = \frac{c + a}{n} = \frac{40 + 144}{194} = .948$$

Similarly, if p_2 represents the proportion who agree that screening mammography is important, then

$$p_2 = \frac{b + a}{n} = \frac{2 + 144}{194} = .752$$

so that

$$p_1 - p_2 = .196$$

In general terms the difference between the two proportions is

$$p_1 - p_2 = \frac{c+a}{n} - \frac{b+a}{n} = \frac{c-b}{n}$$

Here a is the number of tied pairs that contribute to both proportions (in our example, the number who think both are important), and n is the total number of pairs of observations.

The standard error of the difference between the two proportions is given by

$$se(p_1 - p_2) = \frac{1}{n} \sqrt{(c+b) - \frac{(c-b)^2}{n}} = \frac{1}{194} \sqrt{(40+2) - \frac{(40-2)^2}{194}} = .0303$$

The 95% confidence interval of the difference in proportions is

$$(p_1 - p_2) \pm z_{\alpha/2} se(p_1 - p_2) = .196 \pm 1.96 \times .0303 = (.137, .225)$$

9.6 Measures of Association

The choice of a statistical test to describe strength of relationships between variables will largely depend on the level of measurement. Statistical measures of strength of relationship are called either coefficient of association or coefficient of correlation. The former term is generally used when the variables are nominally measured and displayed in a contingency table. The latter term is generally used for variables measured on at least interval (numerical) scale. For example, if we compute **phi** (discussed below) for two nominally measured variables (say for example, sex and eye color), we call **phi** a **measure of association**. On the other hand, if we compute r (discussed in an earlier chapter) for the numerical variables (say age and BMI) we call r a **coefficient of correlation**. In the following sections we will discuss these measures in turn starting with the measures of association first.

9.6.1 Chi Square Based Measures of Association

(a) Phi Coefficient

Phi (ϕ) coefficient is probably the most popular coefficient of association for 2×2 contingency tables. This coefficient is also called four-fold point coefficient. The formula for computing ϕ is

$$\phi = \sqrt{\frac{\chi^2}{n}}$$

Phi (ϕ) coefficient ranges from 0 to 1.0 and attempts to correct chi-squared proportionately to n , the total number of cases. The coefficient does not suggest as to whether one variable causes the other, nor is there an indication of the direction of the relationship. As you can see, the computation of ϕ is straightforward once we have the contingency table.

(b) Cramer's V

Cramer's V is a modification of ϕ for larger tables and has a range upto 1.0 for table of any dimension. It is computed as

$$V = \sqrt{\frac{\chi^2}{n(k-1)}}$$

where k is the lesser number of rows or columns. For a better understanding of the above formula, we rewrite it as

$$V = \sqrt{\frac{\chi^2}{n \text{ minimum } (r-1 \text{ or } c-1)}}$$

(c) Contingency Coefficient

The contingency coefficient was developed for use with square tables having more than two rows and columns, for example, 3×3 or 4×4 tables. It is computed as

$$C = \sqrt{\frac{\chi^2}{\chi^2 + n}}$$

The contingency coefficient C equals 0 when the variables are independent, its maximum value is always less than 1 and is determined by the number of rows and columns in the table. The maximum value for a square table (2×2 or 3×3) is calculated using the equation

$$\sqrt{\frac{(k-1)}{k}}$$

where k equals the number of rows or columns, whichever is less. For a 2×2 table, the maximum value would be attained with $k=2$:

$$\sqrt{\frac{(k-1)}{k}} = \sqrt{\frac{(2-1)}{2}} = .707$$

For a 3×3 table, the maximum value would be .816.

A disadvantage of the contingency coefficient when the variables are neither independent nor perfectly related, is the difficulty of comparing it for tables of unequal size, that is when r is not equal to c . The fluctuations of C 's maximum value has led to the use of the Cramer's V , which can be used with both square and non-square tables of any dimension. V ranges from 0 when the variables are independent to 1, when they are perfectly related.

As with C and V , ϕ only requires nominal level data and provides us with a measure of the strength but not the direction of association, since the sign of ϕ will always be positive.

Cramer's V is the most versatile of the three measures of association since its range is always 0 to 1, and it can be used for a table of any dimension.

Example 9.36: The accompanying table shows the relationship between mothers' educational attainment and degree of malnutrition among 150 under-five children of a city. Compute C , V , and ϕ and interpret the results.

Degree of malnutrition	Educational attainment of mothers			Total
	Low	Moderate	High	
Low	5	10	35	50
Moderate	15	30	5	50
High	30	10	10	50
Total	50	50	50	150

The χ^2 value was computed to be 66, which is highly significant ($p < .001$). Based on this value, we compute other measures:

$$\phi = \sqrt{\frac{\chi^2}{n}} = \sqrt{\frac{66}{150}} = .66$$

$$C = \sqrt{\frac{\chi^2}{\chi^2 + n}} = \sqrt{\frac{66}{66 + 150}} = .55$$

$$V = \sqrt{\frac{\chi^2}{n \times \min(r-1 \text{ or } c-1)}} = \sqrt{\frac{66}{150 \times (3-1)}} = .47$$

The contingency coefficient $C = .55$ leads to conclude that there is a moderately strong relationship between educational attainment and malnutrition, since the maximum value for C in a 3×3 table is .816. An examination of the data in the table also shows that the respondents who have completed higher education tend to have children with low degree of malnutrition. While V ranges between 0 and 1, $V = .47$ indicates a moderate relationship between the variables under investigation.

With ordinal data, there are several alternative methods of measuring the association. Among others, **gamma**, **lamda**, **Kendal's tau**, **Somer's d** are a few of them, which are less frequently used methods of measuring the association in contingency tables and hence we skip to some frequently used methods.

The chi square test tells us whether there exists a relationship between two variables, but it does not tell us what the relationship is. This is achieved through correlation analysis. Such analysis shows us how to determine both the nature and strength of relationship between two variables. The **Spearman's rho (r_s) correlation** is a popular and widely used ordinal measure. rho correlates ranks between two ordered variables.

When both variables are measured on a numerical scale (e.g. weight, height), the most commonly used measure of the association is the **Pearson's product moment correlation coefficient (r)**. These two measures have been discussed in an earlier chapter.

9.7 Relative Risk and Odds Ratio

Case-control studies are conducted to identify risk factors for diseases or conditions. If a difference is observed between cases and controls with regard to a specific variable (risk factor), a chi-squared test can be performed to determine if this difference is statistically significant. However, when this test is performed, the strength of the relationship between the two variables (having or not having the risk factor and having and not having the disease) is not measured. Therefore one needs a measure of association. The **relative risk** is such a measure.

In case-control studies, the relative risk often cannot be calculated because the incidence of the disease in the total population, from which the sample is drawn, is not measured. It is however possible to estimate the relative risk by what is known as **odds ratio**. We discuss these two measures and illustrate the computations by a few examples.

9.7.1 Relative Risk

In some applications it may be preferred to measure the association in terms of the ratio of the proportions. Consider the accompanying contingency table:

Group	Response categories		Total
	A	B	
Group 1	a	b	n_1
Group 2	c	d	n_2

The relative risk (R) is calculated as

$$R = \frac{a/n_1}{c/n_2}$$

In cohort studies, when proportions are of those developing disease during a defined follow-up period, the ratio may be referred to as the **relative rate**. In cross-sectional studies, the ratio may be a **prevalence ratio**.

The standard error is obtained of $\ln(R)$ (where $\ln$ is the natural logarithm to base e), and is given by

$$se(\ln R) = \sqrt{\frac{1}{a} - \frac{1}{n_1} + \frac{1}{c} - \frac{1}{n_2}}$$

The confidence interval is calculated in terms of $\ln(R)$, and then the limits are converted to limits for R by taking the exponential (antilog of natural logarithms).

Example 9.37: In a study of contraceptive use in a district, the possible association between the use of oral pills and whether the women were urban or rural based were investigated. The results were

Group	Pill use		Total
	Yes	No	
Urban women	66	6	72
Rural women	22	85	107

Chi-square=87.1

The relative risk (R) is

$$R = \frac{a/n_1}{c/n_2} = \frac{66/72}{22/107} = 4.46$$

and

$$l_n(R) = 1.49$$

The standard error of $l_n(R)$ is given by

$$se(l_n R) = \sqrt{\frac{1}{66} - \frac{1}{72} + \frac{1}{22} - \frac{1}{107}} = .19$$

Hence the 95% confidence interval for $l_n R$ is

$$1.49 \pm 1.96 \times .19 = (1.12, 1.86)$$

Hence the 95% confidence interval for R is

$$(e^{1.12}, e^{1.86}) = (3.06, 6.42)$$

The chi squared value is 87.1 ($p < .0001$). The difference is highly significant. There is thus very strong evidence that urban women are more likely to use oral pills than the rural women.

9.7.2 Odds Ratio

In case-control studies, relative risk can be approximated fairly by what is known as **odds ratio**. However, this is true if the following two conditions are satisfied:

- The disease has a low incidence rate in the total population
- The control group is representative of the total population.

To illustrate how an odds ratio is computed, consider the following table:

Groups	Risk factor	
	Present (+)	Absent (-)
Cases	a	c
Control	b	d

The odds ratio (O_R) is estimated from a table of above type as follows:

$$O_R = \frac{\text{Odds of cases who have risk factors}}{\text{Odds of control who have risk factors}} = \frac{a/c}{b/d} = \frac{ad}{bc}$$

Calculation of odds ratio is not recommended if the χ^2 value shows that the difference between the cases and the controls is not statistically significant.

The standard error of the $l_n(O_R)$ is computed as follows

$$se(l_n O_R) = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$$

This standard error then can be used to obtain confidence interval for the O_R .

Example 9.38 A case-control study to investigate the association of breast cancer with the use of oral contraception (OC) gave the following data:

Groups	Use of OC	
	Used	Not used
Cases	67	73
Control	118	161

$$O_R = \frac{ad}{bc} = \frac{67 \times 161}{118 \times 73} = 1.25$$

and

$$l_n(O_R) = .225$$

The standard error of $l_n(O_R)$ is

$$se(l_n O_R) = \sqrt{\frac{1}{67} + \frac{1}{118} + \frac{1}{73} + \frac{1}{161}} = .21$$

Hence the 95% confidence interval for $l_n(O_R)$ is

$$.225 \pm 1.96 \times .21 = -.19 \text{ to } .64$$

Exponentiating, we obtain the confidence interval for O_R :

$$(e^{-.19}, e^{.64}) = (.83, 1.90)$$

The odds ratio shows that the contraceptive users are 25% more likely to run the risk of developing cancer than those who do not use. The difference is not however statistically significant as shown by the confidence interval (the central value of the odds ratio, which is 1, lies in the range .83–1.90). This can also be shown by computing the chi-squared

value. That is to say that there is no statistical evidence that use of contraceptives and breast cancer are associated.

EXERCISES 9

1. What is a statistical hypothesis? Why do we need to test a stated hypothesis? Compare and contrast null hypothesis and alternative hypothesis with examples.
2. What is meant by level of significance? How is it determined in testing a statistical hypothesis? Define critical value, critical region, acceptance region and p -value.
3. Distinguish between a type I error and a type II error with examples and that between one-tailed test and two-tailed tests.
4. Describe in brief the rationale behind testing a statistical hypothesis. Enumerate the steps involved in such testing procedure.
5. Distinguish between a normal test and a t test. What are the assumptions underlying a paired t test? Give examples to illustrate the difference.
6. What hypothesis-testing procedure would you use in the following situations:
 - (a) An inspector classifies outputs as defective and non-defective. On the basis of data, he tests the hypothesis that output is not related to machine type.
 - (b) A superstore has 24 salespersons. The test is designed to evaluate whether their sales performance is unchanged or has improved following a training program.
 - (c) A company manufactures automobiles at two different facilities/ It is intended to know if the gas mileage is the same for vehicles from both facilities. The sample size in each case is 50.
 - (d) A social researcher classifies women as working mothers and non-working mothers to see if there is any significant difference in their breast-feeding duration. For this purpose he selects 10 working mothers and 15 non-working mothers.
 - (e) It is suspected that malnutrition is more prevalent among the rural children than among the urban children. Fifty children were randomly chosen from each of the two areas. A test is to be performed to examine the suspicion.
7. A postmaster general claims that the waiting time to serve a customer in a queue is normally distributed with a mean of 3 minutes and a standard

deviation of 1 minute. An inspector found in a sample of 50 customers that the mean waiting time was 6.2 with a standard deviation of 0.5 minutes. At the 0.05 significance, can we conclude that the mean waiting time is less than 3 minutes?

8. The mean length of a small counterbalance bar is 43 millimeters. The production supervisor is concerned that the adjustments of the machine producing the bars have changed. He asks the Engineering Department to investigate. The Engineering Department selects 12 bars and measures each. The results are reported below in millimeters:

42 39 42 45 43 40 39 41 40 42 43 42

Is it reasonable to conclude that there has been a change in the mean length of the bars? Use the 0.05 level of significance.

9. A drug manufacturing company claims that its newly developed medicine is at least 90 percent effective in curing a certain disease. In a random sample 400 people who had the disease, 320 people were cured by using the medicine. Should we conclude that the company's claim is legitimate? Use 0.01 level of significance.
10. A random sample of 8 cigarettes of certain brand has an average tar content of 18.6 milligrams and standard deviation of 2.4 milligrams. The manufacturer, however, claims that the average tar content does not exceed 17.5 milligrams. Use a 0.01 level of significance to test the manufacturer's claim.
11. To test the effectiveness of a pain-relieving drug, 80 patients at a clinic are given a pill containing the drug, while 80 others are given a placebo not containing any drug. If 52 in the first group and 38 in the second group report a positive effect, what can we conclude at 1% level about the difference between the corresponding true proportions?
12. The coefficient of correlation obtained from a random sample of 18 pairs of observations is 0.45. Test the hypothesis at 5% level of significance that the sample has come from a population with zero correlation.
13. The manufacturer of a spot remover claims that his product removes at least 80% of all spots. What can we conclude about his claim at the level of significance 0.01 if the spot remover removed only 154 of 200 spots chosen at random from spots on cloths brought to a dry cleaner?
14. A large automobile manufacturing company is trying to decide whether to purchase Goodyear tire or Bridgestone tire for its new model automobiles. To help at a decision an experiment is conducted using 12 liters of each brand. The tires were run until they were out. The results were as follows:

Goodyear brand: mean=37900 km, SD=5100 km

Bridgestone brand: mean=39800 km, SD=5900 km

Test the hypothesis at 5% level of significance that there is no difference in the two brands of tire.

15. A research study indicates that only 45% children in Bangladesh continue their studies beyond secondary level. A random sample of 200 students shows that 110 students continued their education beyond secondary level. Has there been a significant increase in the proportion of students continuing their studies beyond secondary level? Test at the 5% level. Construct a 95% confidence interval.
16. A machine is found to produce 16 defective items in a sample of 500. When overhauled, the same machine produces 5 defective items in a randomly selected day's run of 100 items. Has the machine improved as a result of overhauling? Test at 5% level of significance?
17. The blood glucose levels of 10 randomly selected persons before (BBF) and after breakfast (ABF) were measured. The results were as follows:

BBF : 5.4 6.1 4.5 4.1 5.2 5.7 2.8 5.5 5.1 6.1

ABF: 7.2 8.1 5.5 6.2 8.1 5.9 5.1 6.1 6.4 8.4

Use an appropriate test statistic to test if the differences in the glucose level are significant at 5% level.

18. The managements of city shopping malls plan to study the customers' shopping habits. From some earlier studies, the managements are under the impression that a typical shopper spends 0.75 hours at the mall on average with a standard deviation of 0.10. Recently the managements added some specialty restaurants designed to keep shoppers in the mall longer. A consulting firm has been hired to evaluate the effects of the added advantages. A sample of 45 shoppers by the firm revealed that the mean time spent in the mall is 0.80 hours. Determine if the mean time spent in the mall has increased from 0.75 hours? Test at 5% level of significance.
19. The government recently granted funds for a special program designed to reduce crime in high crime-prone areas, A study of the results of the program in 10 crime-prone areas of a city yielded the following results:

	Number of Crimes by Area									
	A	B	C	D	E	F	G	H	I	J
Before	14	7	4	5	8	17	12	8	10	7
After	2	3	4	5	5	9	5	4	5	4

Has there been a decrease in the number of crimes since the initiation of the program? Use 0.01 level of significance and compute p -value.

20. The manager of a computer software company wishes to study the number of

hours top executives spend at their computer terminals by type of industry. The manager selected a sample of five executives from each of three industries. At the 0.05 significance level, can the manager conclude there is a difference in the mean number of hours spent at a terminal per week by industry? The accompanying table shows the necessary data.

Banking	Retail	Insurance
12	8	10
10	8	8
10	8	6
12	8	8
10	10	10

- 21 A case-control study was carried out in a health centre to determine whether breast-feeding protects children aged 12–23 months against malnutrition. The following results were obtained.

	Malnourished (cases)	Well nourished (controls)
Brest-fed	100	120
Not breast-fed	100	76

Determine if breast-feeding has any effect on the malnourishment (use 5% level of significance)

- 22 Refer to Q 21 above. It was suspected later that age is a confounding variable, because
- Children aged 12–17 months are more often malnourished than those aged 18–23.
 - More children aged 12–17 months are breast-fed than those aged 18–23 months.

How do you analyze the data to take into consideration the above factors if you have the following splits of the data?

	Children aged 12–17		Children aged 18–23	
Breast-feeding status	Malnourished (cases)	Well nourished (controls)	Malnourished (cases)	Well nourished (controls)
Brest-fed	70	58	60	52
Not breast-fed	50	22	40	40